SYSTEMBOOM INTELLIGENCE ENGINE — MASTER BLUEPRINT

CMI · HEIM · UHI | Edition 1.1 | Owner review | 31 October 2026

Specification only. No database implementation or code modification authorized.

01 | Executive mandate and document controls

Status: MASTER ARCHITECTURAL BLUEPRINT — OWNER REVIEW / IMPLEMENTATION NOT AUTHORIZED BY THIS DOCUMENT. Edition 1.1 · 31 October 2026.

Product: One SYSTEMBOOM Intelligence Engine (SIE), not three separate apps or databases. CMI = media perception; HEIM = memory, retrieval, knowledge; UHI = contextual reasoning, decisions, specialist assistance. An Intelligence Gateway handles authorization and application-specific requests.

Mission: An authorized user can import media once, retain truthful evidence privately, organize it through either direct-category or discovery workflows, reuse validated observations, correct the system, and retrieve connected experiences over time.

This blueprint consolidates product decisions and proposed design; it does not attest that all described capabilities exist in the repository. Baseline: P1 vision/classifier/auto-category developer-reported implemented, owner tested birthday, construction, override and ambiguous cake. Remaining P1 scope and regression not fully owner accepted. HEIM, CMI cross-application service and persistent graph remain future.

Control: Master requirements (REQ-), decisions (DEC-), tests (TST-), phase gates (GATE-) are independently traceable. Any altered assumptions require explicit owner approval and a revision record.

02 | System boundaries and naming

SIE includes (a) CMI universal media perception, media identity, metadata capture and possible media relationships; (b) HEIM observations, cache, user feedback, personalization, knowledge graph, vector retrieval and lifecycle; (c) UHI category assessment, event interpretation, specialist enrichment, user-facing assistance and later agent capabilities. The Gateway enforces permissions, entry context, rate limits, trace IDs, contract versions and safe output filtering.

There is ONE canonical media registry and ONE canonical Human Record model reused across applications, not a parallel HEIM media library. A single asset may link to multiple authorized records, but does not automatically imply a single event. Social publishedAt, media capturedAt, event happenedAt, and record createdAt remain distinct. Raw capture metadata and internal AI observations do not become public merely because the media is shared.

Scope interfaces: Social Composer, Circle of Life, Photo, Video, specialist worlds, future Chat agent, future whole-life timeline. Full video temporal analysis, face identification, automatic global model training, real clinical

interpretation, live external services and agentic posting are future gated expansions—not initial P2 implementation.

03 | User experiences and measurable outcomes

Journey A—Social discovery: select/import media → authorize processing → observe each asset → retrieve HEIM → classify all seven candidates if no authoritative choice → auto-select only reliably eligible case; otherwise show alternatives or neutral fallback → human confirm/edit → existing specialist → private Human Record → optionally share.

Journey B—Circle direct entry: Circle → date → one of seven worlds → upload → preserve user-chosen category → cache/observe → skip general category chooser and general auto-select → run chosen specialist → confirmed record and timeline projection. No silent category switch; mismatches may be shown as optional, nonblocking advice only.

Journey C—Media library reuse: authorized open → reuse compatible observation; no new DeepSeek call on valid cache hit → expose relevant descriptions and links, never promote unconfirmed event identity to fact.

Journey D—Bulk upload: original file identities, timestamps, places, per-asset observations; potential shared events proposed with supporting provenance; grouping approval, no duplicates.

Journey E—Correction: AI suggests Meal for birthday → user selects Life Moment → append feedback with event/asset/decision linkage → future retrieval may prioritize similar verified preferences, without turning all cakes into moments.

Success outcomes: lower redundant model calls, accurate classifications, reliable manual override, robust privacy, accessible mobile flows, coherent cross-world links, versioned evidence and measurable quality.

04 | Architecture and processing flow

Clients → SIE Gateway (authn/authz, capability, consent, origin, trace) → Media registry / asset ingestion → metadata extractor + CMI vision adapter → HEIM versioned observation store + cache → HEIM knowledge/personal retrieval → UHI experience hypothesis and seven-category classifier → separate eligibility/decision policy → category-specific specialist → canonical Human Record. Feedback events return to HEIM.

Provider adapters are replaceable: DeepSeek today, other services later. Never bind persistent record types to vendor-specific JSON. Validate provider output at runtime and normalize. Preserve raw response only under explicit restricted logging/retention policy (prefer not to persist raw response).

Pass 1 remains category-neutral (objects, scene, actions, OCR, primary subject, limitations, provenance). Pass 2 outputs candidate experiences and category rankings. Decision Policy is a separate gate which may auto-select, suggest, ask or no-match. Specialist enrichment follows category authority.

05 | Entry context, category authority and no-match

Three modes: DISCOVER (Social with no explicit category), ENRICH (Circle/direct world with category), REUSE (authorized media read). Request must carry entry context, media IDs, selected category where appropriate, provenance, policy version, and an authenticated capability; never accept arbitrary owner ID from client as authoritative.

Priority: user-selected > direct-entry specialist > eligible AI-selected > default placeholder. Explicit human selection locks AI auto-selection in the current editing session; late results cannot mutate category or cleared/confirmed/corrected specialist fields. Direct entry bypasses category classifier while retaining Pass 1, HEIM and specialist processing.

Life Moment is both a genuine experience category and neutral save fallback; distinguish noMatch from confident Life Moment recognition. Empty chair → neutral fallback, not claim an event. Mixed post asset categories → do not force unless validated group context and eligible policy. Safe retry must not publish, change visibility or delete media.

06 | Central media identity and provenance

Registry binds canonical asset ID, tenant/user authorization, original encrypted location, content SHA-256, MIME, size, derivatives lineage, created/ingested timestamps and source, plus metadata provenance. Exact-byte duplicates can reuse compatible observations only within authorized scope. Perceptual hashes and embeddings generate candidates, never merge identity or user records by themselves.

Metadata: EXIF capture date may be incorrect/missing; store value, precision, source, confidence, time zone if available; GPS requires separate location-processing permission; user can accept/adjust/ignore a location suggestion; original private metadata need not be deleted when Social location sharing is disabled. Public place is a separately approved field.

Media relationships: possible-same-event, possible-same-place, same-asset-derivative, attached-to-record and explicitly-confirmed-event edges. Provenance and confidence are mandatory, no automatic identity inference of people. Video starts with metadata and explicit supported frame sampling only; do not claim full video understanding.

07 | HEIM observation cache and invalidation

Key = authorized scope + original content hash + normalized input variant + provider/model + prompt version + observation schema + preprocessing version + analysis-policy version. Identical bytes do not imply cross-user sharing rights. Cache flow: request → validate capability → acquire single-flight/lease → lookup valid COMPLETE entry → return observation if authorized → otherwise analyze → validate → atomically persist → release; failed/transient results never treated as valid observations.

Use stable deduplication for concurrent same-key calls; lock/lease expiration, retry/backoff, crash recovery, quota/timeout, negative-caching only for bounded permanent invalid-input failures. Exactly-once provider billing cannot be guaranteed across crashes; target no repeat calls for successful compatible cached results. Version bump invalidates or triggers safe reanalysis while old observations remain auditable. Handle edit/resize/crop derivatives explicitly.

Cache hit never reuses an old category decision automatically: Pass 2 and policy may be re-run using current user context and knowledge version without re-calling Pass 1.

08 | HEIM correction and preference memory

Persist append-only decision + feedback ledger: prediction, evidence refs, policy/model/knowledge versions, selected category, action type (accept/override/clear/adjust), source (explicit user/AI/default/direct-entry), asset/group/record, timestamps, consent and authorizing principal. Do not infer positive labels from absence of correction.

On explicit override: persist current authoritative choice first; append feedback idempotently with decision ID and client action ID; update personal-preference projection asynchronously with bounded evidence, context features and recency; preserve original error for evaluation. Separate per-field correction/clear events and never override protected data.

Use comparable experience evidence and user preference only when scope/context match; never globalize private feedback automatically. Guard against training on the model's own predictions, one-shot overfitting, feedback manipulation, contradictory user history, and sensitive-category learning without consent. Provide review, export, correction, deletion/retention controls.

09 | Typed Knowledge Graph, not 10,000 conditionals

Nodes: observable object, action, setting/scene, OCR phrase concept, document kind, event/experience, record category, contextual qualifier. Typed directed edges: INDICATES, COMMON_AT, CAN_OCCUR_DURING, SUPPORTS, CONTRADICTS, REQUIRES_CONTEXT, PRIMARY_EXPERIENCE_OF, ALTERNATIVE_LENS, IS_A. Edges carry strength/quality, source/provenance, review status, version, context constraints and validity window. The graph is not a set of automatic deterministic category assignments.

Example: OCR phrase HAPPY BIRTHDAY → indicates birthday celebration; balloons/candles → supports; cake → common_at birthday but also food; birthday celebration → primary experience Life Moment; cake-alone → may support Meal. Worker → CAN_OCCUR_DURING construction, but not proof the user's activity. Whiteboard in office requires agenda/discussion evidence before Meeting. Hospital bed supports care-setting, never diagnoses.

Retrieval: normalize observed concepts → query bounded candidate subgraph → check evidence dependence and counter-evidence → retrieve scoped correction examples → calculate experience hypotheses → category ranking → calibrated policy. Repeated model-derived clues from one image must not be counted as independent proofs. Graph writes require reviewed provenance, tests and versioned promotion. Start relational in PostgreSQL; graph DB only after measured need.

10 | UHI reasoning, confidence and decision policy

Keep separate values: observation quality, experience support, category suitability, calibrated prediction reliability, user intent and action eligibility. Provider 0.98 is a score, not a validated 98% probability. Thresholds chosen against held-out real data and tracked by category/risk. Strong birthday with OCR+scene+objects supports Life Moment auto-selection if no explicit human choice and policy passes. Close-up birthday cake with competing supported meanings → ask. Empty chair → no strong specialist match.

Policy must account for source overlap, alternative margin, per-category safety, image type, locale, direct entry, multi-asset conflict, Smart Assist OFF, stale jobs, sensitive Health and unsupported fields. Health suggestion must never be confused with diagnosis or clinical care. Decision log stores why a category was considered eligible, which policy ran, and whether category was AI-selected vs human-selected.

Specialists remain separate: Meal, Activity, Health, Problem, Project, Meeting, Life Moment. Provisional suggestions must use honest wording and be editable; no inferred Project Goal, Meeting Purpose, diagnoses, ages, relationships, attendance or ownership without evidence.

11 | Video, text, documents and future modalities

P2 focuses on still-image memory integration, contracts and architecture for video. Future CMI modality adapters support video sampled frames/temporal segments, audio transcription with consent, imported documents, text/chat and structured device data, each with modality-specific observation types and provenance. Do not pretend a thumbnail analyzes an entire video.

Cross-modal concept and event retrieval must be permission filtered and scoped to canonical assets. Media organizer uses the same observation cache without hijacking classification or forcing record creation. Future Chat agent acts under separate explicit capability grants; no automatic external actions or public sharing.

12 | Data model and database mapping

Proposed PostgreSQL foundation: existing canonical users/media/Human Records remain authoritative. Add namespace heim with observation_runs, evidence_items, observation_cache, category_decisions, correction_events, personal_preference_projection, concept_nodes, concept_edges, knowledge_releases, embeddings, media_event_links, consent_policies and evaluation_cases (final names adjusted after P2-A audit). Avoid copying existing entities. UUID FKs, tenant/user access rules, JSONB for versioned observations, vector index if measured search requires it.

No migrations until audit verifies existing IDs, ORM, persistence, session scope, deploy topology and privacy model. Local SQLite can model most relations but pgvector/RLS semantics require PostgreSQL testing. Encrypt source files and high-risk OCR; separate private intelligence serialization from public post allowlist. Make schema migratable and reversible; preserve provenance and correction immutability.

Detailed DDL from the existing HEIM P2 package is a reference proposal, not an authoritative migration: reconcile differences in names and ownership before implementation.

13 | TypeScript and API contracts

Define: MediaIntelligenceMode DISCOVER|ENRICH|REUSE; EntryContext social_composer|circle_of_life|specialist|media_library|future_agent; Category life_moment|meal|activity|health|problem|project|meeting; SelectionSource default|ai|user|direct_entry.

Core request: {requestId, assetIds[], mode, entryContext, selectedCategory?, selectionSource?, recordId?, locale?, privacyCapabilityId?, clientActionId?}. Server binds principal, checks access and validates mode/category combinations.

Core result: {analysisStatus, observationsByAsset, experienceCandidates?, categoryDecision?, selectionState, specialistSuggestions?, privacySafeSummary, provenance, cacheStatus, warnings[]}. DISCOVER may provide categoryDecision; ENRICH must not reselect category; REUSE cannot infer consent.

Interfaces and strict runtime schemas cover VisionObservation, EvidenceRef, ExperienceHypothesis, CategoryDecision, CorrectionEvent, KnowledgeConcept/Relation, RetrievalEnvelope, CacheLookup and PolicyDecision. P2-A audits current TypeScript names before adoption; do not introduce duplicate category enums or sidecar authority state.

14 | Gateway, authorization, privacy and consent

Gateway authenticates every request, binds tenant/owner server-side, authorizes asset and record, checks action purpose and scoped consent, isolates provider secrets and logs structured metrics without raw contents. Public Social response uses strict field allowlist; internal visualObservation, OCR, confidence, graph evidence, embeddings, health/location metadata and feedback do not pass into public feed/search by default. Hiding a UI key alone is insufficient.

Separate toggles: analyze media, extract GPS, suggest private place, attach/share chosen Social location, contribute data to aggregate learning. Enable/disable must have documented effects; revocation stops future processing and triggers retention/deletion workflows where applicable. Children and medical content warrant stricter treatment, access, retention and diagnostics controls.

Private, family-shared and global knowledge scopes cannot implicitly inherit permissions. RLS and server authorization defense-in-depth; storage encryption; sensitive logs redacted; breach/backup/restore planning required before production.

15 | Quality, observability and economics

Metrics: top-1 accuracy/top-3 recall by category, auto-select precision and coverage, misclassification by subject ambiguity, manual override rate, stale-response defects, unsupported specialist fills, per-asset cache hit/miss, duplicate provider call rate, p50/p95 latency, provider cost, knowledge retrieval latency, privacy leak count and mobile completion. Baseline measured before threshold commitments.

Maintain golden evaluation corpus with owner-approved representative real/synthetic/license-clean examples, splits per event/category/person/device/locale and negative cases. Avoid leakage from training into evaluation; monitor drift. A corrected feedback event is not automatically ground truth for global training. Shadow evaluation before changing graph release or policy; promote only with reproducible improvement and rollback.

Handle outage modes: manual Composer usable, cached authorized observations reused if valid, no fabrications, partial observations labeled, specialist error isolated, idempotent save and consistent UI during concurrency.

16 | Delivery phases, owner gates and definition of done

P1 STABILIZATION: finish six owner workflows including mixed media; track bubble-tea Beverage-vs-Snack as separate specialist issue; category override, birthday and construction verified in browser; physical mobile and privacy audit still need evidence. No retroactive claims of total P1 acceptance.

P2-A AUDIT ONLY: inventory current code/data/IDs/AI calls/privacy, reconcile HEIM package and CMI-001/CMI-002, propose schema and minimal vertical slice. Deliver docs, no code/database migration; gate GATE-A owner reviews architecture.

P2-B1 CANONICAL MEDIA + GATEWAY CONTRACTS: adapter and ownership boundary; no replacement of canonical media. GATE-B1 contract tests.

P2-B2 OBSERVATION PERSISTENCE + CACHE: local sanctioned database, identity, versioning, single-flight, privacy; GATE-B2 replay and isolation.

P2-B3 CORRECTION LEDGER: explicit feedback, preference retrieval, authority; GATE-B3 replay and non-overwrite.

P2-B4 KNOWLEDGE GRAPH FOUNDATION: curated concepts/edges/versioned retrieval; GATE-B4 reasoning and provenance tests.

P2-B5 CLASSIFIER INTEGRATION: controlled use of cache, knowledge, preferences across discovery and direct enrichment; GATE-B5 full regression.

P2-B6 HARDENING: performance/privacy/retention/security/migration/backup evaluation; GATE-B6 independent verification and owner acceptance.

P3+: future multimedia temporal reasoning, full event linking, agent functions and training remain separate owner decisions. No automatic commit/push/deploy/production writes.

17 | Stakeholder responsibilities and deliverables

Owner/CEO: approves category policy, privacy defaults, scope and gate acceptance; reviews real experience quality. CTO: reconciles canonical models, performance, architecture and migration. AI lead: model/observation validation and evaluation. Data lead: graph semantics, provenance, cache, feedback. Privacy/security lead: consent, sensitive data access, retention and leakage. UX/accessibility: mobile-first feedback, correction state, no contradictory advice. QA: independent E2E, fixtures and regression gates.

Deliverables: master architecture document; traceable requirements and acceptance register; canonical entity/API/SQL migration plan (after audit); graph seed governance; test evidence; QA report; known risk register; owner decision log; developer handover; runbook and rollback strategy. All code changes local until separately approved.

18 | Risks, open decisions and non-goals

R-01: no canonical persistence in prototype → audit before choosing ORM or migrations. R-02: auto-selection scores uncalibrated → benchmark and per-category gates. R-03: similar media incorrectly merged as one event → hypothesis only until verified. R-04: personal memory leakage to other users/global model → strict scope, no silent promotion. R-05: OCR/health/children location exposure → privacy-by-default and explicit consent. R-06: knowledge graph drift/bias → versioned reviewed graph release. R-07: provider data cost and licensing → inventory vendor data and policies. R-08: changed category stale AI card or overwritten values → authority and race tests.

Owner questions before P2-B: actual database/deployment target; authentication/tenant scope; opt-in granularity for learning and GPS; retention/deletion policy; whether historical private media should be backfilled; graph seed stewardship; exact eligibility thresholds after benchmark; P2 local data fixture policy; release/rollout strategy. No blocking question prevents the read-only P2-A audit.

Non-goals: replacing canonical Human Records; making three separate apps; implementing general facial identity; diagnosing illnesses; auto-publishing; auto-correcting user choices; labeling all media as one event; production migrations under audit authorization.

19 | Representative end-to-end scenarios

Birthday party: multiple cues including HAPPY BIRTHDAY → Life Moment eligible for auto-selection; cake still Meal alternative when relevant; human change persists. Cake close-up: Meal or ask if birthday context; no fabricated party. Construction: Project not incidental Activity; goal empty. Running: Activity. Office whiteboard: not automatically Meeting absent evidence. Meeting agenda: Meeting possible when supported; purpose not invented. Hospital document: Health suggestion, private, no diagnosis. Cracked phone: Problem, no invented cause. Trophy/medal: possible achievement Life Moment or Activity contextual lens. Empty chair: no-match, neutral option.

Circle → Project → media: skip classifier, still observe and enrich; Circle → Meal → video: category fixed and only actually supported video analysis; Smart Assist OFF: no model calls; User selects Meeting after Project auto: Meeting stays, AI card neutral; multi-photo birthday+construction: no forced category; Social post: original capture and internal GPS private, publishedAt independent.

Future timeline: uses canonical event/happenedAt and authorized media links and feedback, does not fabricate events or use Social publishedAt as historical event date.

20 | Acceptance policy and signoff

All acceptance cases are in the companion test register, each with unique ID, phase, component, entry, action and objective assertion. States: NOT RUN / BLOCKED / DEVELOPER-REPORTED PASS / INDEPENDENTLY VERIFIED / OWNER ACCEPTED / FAIL. No 'pass' without logs/evidence. Real-provider, mock, browser emulation and physical-device results tracked separately. Privacy and authorization issues are release blockers.

Go/no-go: P2-A DONE only with evidence-based repository audit, compatibility decisions, risks and phased plan; P2-B starts solely on separate approval. Product release requires no P0 security/privacy defects, quality thresholds supported by a benchmark, E2E current flows and explicit owner signoff. This blueprint is an architectural target, not evidence of completed implementation.

21 | V1.1 reconciliation: audited corrections

The V1.0 CSV contained 82 uniquely numbered but generic requirements (e.g. capability 1) and 162 scenarios with broad, repeated expected-invariant language. Those registers were unsuitable as a developer acceptance baseline. V1.1 supersedes those registers with individually testable requirements and traceable tests. The source V1.0 package is retained unchanged.

V1.1 is a design review baseline, not a verified implementation baseline. Actual identity, media asset tables, authorization, ORM, model API use, and persistent store must be confirmed by the P2-A read-only audit. HEIM SQL remains a reference: never execute before repo compatibility, access policies and local migration review.

Required non-negotiable design gates

- Existing canonical Human Record and media IDs remain authoritative; never create shadow records.

- Social DISCOVER: observe, retrieve, classify, then optionally auto-select. Circle/direct ENRICH: observe, retrieve and specialize without seven-category classifier. REUSE: authorized cached read.

- Evidence is not intent. Explicit user correction overrides all AI selections; raw visual summaries are private, not public post content.

- Video analysis must declare actual frame/time coverage; no claim of complete video understanding based on a thumbnail.

- Successful same-asset compatible Pass 1 observations reuse cache. Provider billing exactly-once is not guaranteed across crash windows; lease/retry semantics must be tested.

- Correction memory is append-only and personal unless separately consented and reviewed for global promotion. No passive publication treated as confirmation.

- Global Knowledge Graph concepts are reviewed/versioned; edges are probabilistic candidate relationships, not proof of personal facts.

- Threshold numbers are provisional until real-image calibrated evaluation; risk-sensitive Health never triggers unauthorized processing.

22 | AI cost and latency policy (mandatory)

Use zero-LLM work first: trusted original-asset hashes, metadata parsing, deterministic validation, access checks, bounded graph lookups, cache hits and rule/policy gates. Only invoke provider when validated cache miss needs visual understanding. Do not send the same bytes twice to separate Vision + classifier calls if observations suffice for a deterministic/classifier pass. Defer embeddings until measurable added value; batch where user latency permits. Log model, token counts if available, latency, cache hit, provider call count and cost estimate; never log OCR or images. Set request limits, one retry budget, timeout/circuit breaker and user fallback. Reclassification under new policy uses stored observation where compatible. A category-level model should never be asked to extract specialist clinical or nutrition facts without authorization. Do not use another model simply to verify a model unless evidence indicates it materially improves calibrated decisions.

23 | Performance, confidence and evidence gates

No arbitrary model-returned 0.95 may be treated as 95% accuracy. Establish a consented fixture evaluation set including: clear birthday, close-up cake, ordinary meal, mixed meeting/work, whiteboard not meeting, hospital context without diagnosis, no-match chair, children, OCR multilingual, camera date/accuracy, GPS privacy, damaged photo, provider mismatch and user overrides. Measure auto-selection precision and coverage per category, number of user corrections, latency p50/p95, per-upload call count/cost, repeat-upload cache hit, and public leakage attempts. Targets must be proposed by engineer and approved by owner after baseline.

24 | Data model and implementation safety

Reference schema: attached original HEIM SQL/TypeScript are candidate contracts, not applied migrations. Require documented FK ownership resolution, row-security policies (or app authorization), tenant isolation, OCR encryption/retention, and public allowlist. Cache key includes original hash, representation, model/prompt/schema/preprocessing/policy scope. Knowledge graph: concepts/edges/evidence/confidence source + release, personal overlay isolated. Correction feedback: actor+decision ID+action id for idempotency, provenance and undo. SQLite may mock non-vector contracts locally but cannot establish pgvector / PostgreSQL RLS parity; mark deviations explicitly.

25 | Audit decisions and controlled roadmap

P2-A: read-only evidence audit, compare existing repo to blueprint and proposed schema, enumerate blockers, produce owner decisions. P2-B1: central identity/contracts; B2: observations/cache; B3: feedback; B4: curated graph; B5: retrieval and classifier integration; B6: privacy, tests, performance. No P2-B authorization from this document; no commit, push or deploy.

26 | V1.1 deliverable manifest

- SIE_REQUIREMENTS_REGISTER_V1.1.csv: independently measurable requirement statements and linked tests.

- SIE_ACCEPTANCE_TEST_REGISTER_V1.1.csv: one direct linked test per requirement plus targeted end-to-end edge cases. Tests are NOT RUN.

- SIE_ARCHITECTURE_AUDIT_V1.1.md: findings, priority corrections, release gate.

- CLAUDE_CODE_P2A_READ_ONLY_AUDIT_PROMPT.md: audit-only developer instructions.

- SIE_INVESTOR_ONE_PAGE.md: short business summary and truthful maturity.

- Original HEIM heim_schema.sql, heim_contracts.ts included as REFERENCE ONLY.

V1.2 — AI provider provenance and local intelligence database

21 | V1.2 final design amendment — AI provider provenance & local intelligence database

SIE-PROV-001 (owner approved): Every generated intelligence artifact SHALL have a traceable producer invocation; this includes Pass 1 observations, Pass 2 rankings, specialist suggestions, embeddings and extracted OCR. The ledger SHALL preserve requested provider/model and actual provider/model as separately evidenced fields; where intermediaries obscure routing, actual provider/model is unknown, never a guessed attribution. The ledger SHALL record prompt-template ID and immutable version or digest, observation schema version, processing/preprocessing version, knowledge and policy versions when applicable, UTC invocation times, status, latency, bounded usage and pricing snapshot. Cache hits produce a cache-access event linked to the original artifact; they SHALL NOT impersonate a new model invocation.

AI switching: Adapters implement the same versioned CMI observation and UHI decision schemas. Provider selection and fallback are configurable and independently testable. Output validation and the human-authority invariant remain unchanged across DeepSeek, OpenAI or future providers. Model revisions may be undisclosed; save the actual model identifier when supplied, otherwise explicitly record unknown.

Developer-local persistence decision: Use a dedicated local PostgreSQL database (suggested systemboom_intelligence_dev) or a dedicated heim schema inside a local PostgreSQL instance, never a production database. PostgreSQL + JSONB is the baseline; pgvector is an optional explicitly gated extension for embeddings and similarity retrieval after measured benefit. Preserve media bytes in authorized media/object storage; store canonical asset references and SHA-256 plus representations in HEIM. Keep canonical SYSTEMBOOM Human Record/asset/user IDs; prototype IDs are bridge-mapped rather than silently declared identical to production IDs. Migrations are separately owner-authorized after P2-A.

Proposed relations (additive to HEIM reference schema): heim.ai_providers, heim.ai_models, heim.ai_invocations, heim.ai_artifact_provenance, heim.ai_pricing_versions. Refer to SIE_PROVIDER_PROVENANCE_REFERENCE.sql for column and constraint design. Include authorization and lifecycle rules before implementation. Do not include API keys, raw OCR or image bytes in the ledger.

Cache compatibility boundary: A successful analysis is reusable only when authorization scope, canonical asset digest, representation/preprocessing version, model/provider and prompt/schema versions satisfy compatibility policy. Enforce cache admission and concurrency controls; provider outage is never recorded as a valid observation. A newer classifier can reuse a compatible older observation but must produce a newly versioned decision.

Cost/quality policy: Prefer authorized cached observations; bound graph retrieval, output tokens and timeouts; use deterministic rules for validation/formatting; invoke a model only for tasks that need semantic inference. Capture provider-reported token counts when present, estimated cost using versioned price metadata, latency, cache hit ratio, and error/override rates. Do not falsely imply every multimodal provider uses identical token metrics. Quality gates prohibit token savings that reduce category or privacy performance without owner approval.

Proof and status: The requirements and test registers now contain 92 requirements and 102 proposed tests (V1.2; V1.1 was 82/92). New tests have NOT RUN status. This is a finalized architecture handover draft for P2-A audit, not an independently verified production implementation. P2-B migration authorization is still withheld.

22 | Controlled specification hierarchy and release gates

SIE V1.2 is the consolidated cross-component blueprint; UHI V0.4 is the historical category/experience engineering source; HEIM P2 is the detailed memory reference. Existing numbering and tests must be reconciled by P2-A, not silently discarded or relabeled as passed. CMI-001 describes the shared media service; CMI-002 direct Circle/specialist ENRICH entry must bypass general category selection while retaining observation, HEIM retrieval and selected specialist intelligence. Social DISCOVER can auto-select only on eligible evidence; explicit human choice always wins.

Gate A: Owner review of architecture and document package. Gate B: P2-A codebase/database read-only audit. Gate C: Owner signs off reconciled repository-specific SQL, privacy controls, migrations, rollback and benchmarks. Gate D: Only then local P2-B implementation; never commit, push, deploy or touch production without explicit authorization.